Nomodynamics: The Mathematics of Self-Amending Law

A founding note

The Treeline Program

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Abstract

Game theory formalized games with fixed rules; Conway's deletion of the *players* gave cellular automata and the Game of Life. The orthogonal deletion—removing the *fixity of the rules*—was never taken, although rules that amend rules are humanity's oldest formal practice. We introduce a minimal, parameter-free mathematical object native to that vacancy: a system whose only substance is laws acting on laws. Its founding ("own-kind") semantics turns out to be exactly solvable, with short theorems that carry immediate institutional readings: fully occupied codes are frozen (*gridlock*); stable codes are always fully blocked (*dead letters*); and on the integer line *the eldest law cannot be repealed*, which makes free-moving law-packets impossible—and on circular codes, which have no eldest law, what looks like a moving packet turns out to be a barber pole. The solvable sector exhibits a small charismatic fauna (colonizers, sunset clauses, welds, refraction, apparent rotations, and an aperiodic avalanche machine locked to powers of two). Letting laws amend *other* laws then settles what motion costs: a tropical monovariant shows that a constitution in which every law amends a single kind—anyone's kind—is motionless in every dimension, window and resolution, while two targets suffice, and the smallest traveller is one placed law. The same generalization splits the stability theorem by resolution convention and produces *balanced constitutions*: codes perpetually active and perfectly unchanging. Dropping instead the tacit assumption that laws are permanent inverts the founding theorem—the very first specimen becomes a glider, and a packet's speed is set by how long a provision survives unrenewed. Finally the system computes: a provision that repeals itself every step is a register, a constitution in the resulting normal form is a synchronous AND-NOT network, Rule 110 runs inside a 24-kind constitution, and every Turing machine compiles into a finite code—so the field is computation-universal, with halting undecidable, while every one of its provisions stands still.

1 The vacancy

Three observations, from a systematic survey of how new mathematical ontologies have historically been found, converge on one empty cell. (i) The deletion grid: von Neumann–Morgenstern formalized 2-player fixed-rule games; puzzles are the 1-player case; Conway's "0-player games" are cellular automata. Along the orthogonal axis—rule fixity—no elementary mathematics exists. (ii) The reduction shadow: computability theory's founding reduction, "without loss of generality the program is fixed," is valid for *what can be computed* (universality lets a fixed machine simulate a self-modifying one) but silently discards the *behavioral* questions: what do small self-amending systems actually do? Nobody looked, because the reduction said one need not. (iii) The practiced verb: constitutions and statutes—self-amending by essence—are the oldest running formal systems, yet "amend" was never objectified the way Church and Turing objectified "compute." The fragments that exist (von Neumann's rule-carrying constructor cells; Suber's game *Nomic*, 1982; self-modifying Petri nets, 1978; algorithmic chemistry) never produced a minimal canonical object, a census, or a theorem.

2 The object

2.1 The founding system

Fix the finite set $K_0 = \{-1, 0, 1\}^3$; its elements are called *kinds*. For a kind $k = (a, b, c)$ the three integers are its *precedent*, its *exception* and its *target* offset. A *placed law* is a pair $(i, k) \in \mathbb{Z} \times K_0$; a *code* is a finite set

$S \subseteq \mathbb{Z} \times K_0$ of placed laws; $\mathcal{S}$ denotes the set of codes. Write

$$\operatorname{occ}(S) = \{\, i \in \mathbb{Z} \;:\; (i,k) \in S \text{ for some } k \in K_0 \,\}$$

for the set of *occupied* positions.

Definition 1 (activity)**.** A placed law $(i,k) \in S$ with $k = (a,b,c)$ is *active in* S if $i + a \in \operatorname{occ}(S)$ and $i + b \notin \operatorname{occ}(S)$. Let $A(S)$ denote the set of placed laws active in S.

Definition 2 (nomic chain)**.** Every $(i,k) \in A(S)$ with $k = (a,b,c)$ *emits a toggle* of the placed law $(i+c,k)$. Let $T(S)$ be the multiset of all emitted toggles and let $T_{\mathrm{odd}}(S) \subseteq \mathbb{Z} \times K_0$ be the set of placed laws occurring in $T(S)$ an odd number of times. The successor of S is

$$\Phi(S) \;=\; S \mathbin{\triangle} T_{\mathrm{odd}}(S) \qquad (\triangle = \text{symmetric difference}),$$

and the *nomic chain* is the discrete dynamical system $(\mathcal{S}, \Phi)$, iterated synchronously.

In words: a law acts when the position named by its precedent offset is occupied and the position named by its exception offset is empty; its action is to insert a law of its own kind at the position named by its target offset if none stands there, and to delete it if one does. Φ is well defined because S is finite, hence so are $A(S)$ and $T(S)$.

Example 3. Let $k = (0,1,1)$ and $S_0 = \{(0,k)\}$, a single law at the origin. Then $\operatorname{occ}(S_0) = \{0\}$; the law is active, since $0+0 \in \operatorname{occ}(S_0)$ and $0+1 \notin \operatorname{occ}(S_0)$; it emits a toggle of $(1,k)$, which lies outside S_0, so $S_1 = \Phi(S_0) = \{(0,k),(1,k)\}$. At the next step the law at 0 is inactive ($0+1$ is now occupied) while the law at 1 is active, so $S_2 = \{(0,k),(1,k),(2,k)\}$, and inductively $S_n = \{(0,k),\ldots,(n,k)\}$: the kind $(0,1,1)$ fills the line rightwards at one position per step, only ever acting at its own frontmost copy. Under the lifetime variant with $\tau = 1$ the same computation gives $S_n = \{(n,k)\}$ instead — the single law walks (§7).

2.2 The general object

The later chapters each replace exactly one clause. A *constitution* consists of a finite set K together with, for every $k \in K$: offsets $a_k, b_k, c_k \in \{-w,\ldots,w\}^d$, a *target set* $T_k \subseteq K$, and a pair of *citations* $g_k, h_k \in K \cup \{*\}$. Codes are now finite subsets of $\mathbb{Z}^d \times K$. For $x \in K \cup \{*\}$ and a position p write

$$S \models x@p \quad :\Longleftrightarrow \quad \begin{cases} p \in \operatorname{occ}(S) & x = *, \\ (p,x) \in S & x \in K, \end{cases}$$

so $*$ reads *some law stands here* and $x \in K$ reads *that particular kind stands here.* Then

$$(i,k) \in A(S) \iff S \models g_k@(i+a_k) \;\text{ and }\; S \not\models h_k@(i+b_k),$$

every active (i,k) emits a toggle of $(i+c_k, t)$ for each $t \in T_k$, and Φ is as in Definition 2. The founding system is the case $d = w = 1$, $K = K_0$, $T_k = \{k\}$ and $g_k = h_k = *$; chapter two varies T_k, chapter three varies (g_k, h_k).

2.3 Variants

Each of the following substitutes for one clause and leaves the rest intact.

- *Rings.* Replace $\mathbb{Z}^d$ by $(\mathbb{Z}/m)^d$.
- *Resolution.* Parity, as above, or *OR*: $\Phi_\vee(S) = S \mathbin{\triangle} \operatorname{supp} T(S)$, where a toggle lands if it is emitted at all. The two coincide whenever no placed law receives two toggles in the same step.
- *Supersession.* An active law enacts its own kind at $i + c_k$ if that position is empty, and otherwise deletes every law standing there.
- *Lifetime* $\tau \in \{1,2,\ldots\} \cup \{\infty\}$. Effects are *enactments* rather than toggles: an active law inserts $(i+c_k, t)$ for each $t \in T_k$, and a placed law survives a step only if it is enacted now or has been enacted within the previous $\tau - 1$ steps. Chapters one to three are the toggle semantics; chapter four is $\tau < \infty$.

There is no board, no background and no external rule table: the only data is which laws stand where, and the rule a law obeys is the law itself. The founding system in particular has no parameters — K_0 has 27 elements and that is the entire specification. All claims below were pre-registered as predictions or risks before the first simulation, then settled by censuses with machine-checked certificates (∼15.3 million certified runs for the no-go results; complete enumerations where stated); `verify.py` in the archive re-derives every named claim from the definitions above.

3 First theorems: the solvable sector

Theorem 4 (Gridlock)**.** *In a fully occupied region every law's vacancy guard fails; a solid code is frozen. All dynamics is surface dynamics.*

Lemma 5 (Single Author)**.** *The only law that can ever flip kind $k = (a, b, c)$ at position j is the kind-k law at $j - c$. Hence parity and idempotent ("OR") resolution coincide identically, in every dimension, and per-kind dynamics is occupancy-modulated linear algebra over $\mathbb{F}_2$.*

Theorem 6 (Dead Letter)**.** *A code is a fixed point iff every law in it is blocked. In particular* balanced constitutions*—codes in perpetual self-cancelling activity—do not exist: stability is always gridlock.*

Theorem 7 (Anchor)**.** *On $\mathbb{Z}$ (any window, any dimension, any guard predicate, either resolution semantics), own-kind toggles of kind t land only at t's own offset c_t; consequently the extremal law on the trailing side of a finite code is never targeted:* the eldest law cannot be repealed. *No finite code satisfies $\Phi^p(S) = \sigma^d(S)$ with $d \neq 0$: free gliders are impossible.*

Corollary 8 (Ring rotors—and what they are not)**.** *On $\mathbb{Z}/m$ there is no eldest law, and codes appear that coincide with their own rotation: three laws of the single kind $(0, 1, -1)$ at cells $\{1, 2, 5\}$ of $\mathbb{Z}/6$ satisfy $\Phi(S) = \mathrm{rot}_3(S)$ (Fig. 2), and the family runs over every even $m \geq 6$—the block word forces $m = 2\gamma + 4$ with hop $m/2$, so the evenness is a theorem, not a survey.* But this is not transport. *Window-1 laws move information at most one cell per step, so a rotation by $m/2$ in a single step cannot be carrying anything: exactly two cells change—a repeal at 1, an enactment at 4—and the new state merely happens to equal the old one rotated (the seed has no rotational symmetry, so 3 is the only rotation relating them). Every own-kind ring rotor fails this light-cone test, and releasing* 19,074 *of them onto $\mathbb{Z}$ yields zero gliders.* Own-kind nomodynamics contains no transport at all: not on the line, and not on the circle. What a ring supplies is apparent rotation—a barber pole.

Remark 9. The lemma explains the observed clockwork: every exact cycle found across ∼140,000 seeds has period a power of two, and single-kind ring codes (the "Sunset Parliament") attain exact maximal periods 15, 63, 341 at resonant circumferences $m \equiv 2 \pmod 4$—orders of $\mathbb{F}_2$-polynomials governing constitutional cycles.

4 Fauna of the solvable sector

The one-dimensional chain is a calculus of fronts (Fig. 1): *colonizers* (2 of the 27 kinds) fill the line at speed 1; the *sunset clause* $(0, -1, 1)$ forever enacts a subordinate its own presence then blocks (period 2); *sunset codes* dissolve from their newest end; colliding fronts *weld*; porous codes support one-way *conversion waves* (speed 2/3); walls facing a front sort into opaque (19/27), pulsing (3/27), and effectively transparent (5/27) classes. In two dimensions a *ray-confinement theorem* pins each kind to one axis ray (quadrants never fill), and the fauna adds 239 certified half-turn *pinwheel rotors*, Pascal-column growers with $|S_t| = 2^{\mathrm{popcount}(t)}$, and the Jubilee Code (Fig. 2, right). Finite rings admit an exact fixed-point count (transfer matrix, brute-force verified), a predecessor algebra with Garden-of-Eden fraction peaking ≈ 33% at mid-density, and a unique minimal taut architecture (the two-law *mutual-veto* pair).

5 Cross-amendment: what motion costs

Own-kind amendment is the tame simplification; real statutes amend other statutes. Generalize minimally: a *constitution* is a finite kind set K, each $k \in K$ carrying a rule (a_k, b_k, c_k) and a *target set* $T_k \subseteq K$; an active law of kind k at i toggles every kind of T_k at $i + c_k$. Own-kind is $T_k = \{k\}$. Nothing external is added: out-degree 1 makes the amendment digraph a functional graph, so a kind is a pointed finite graph with a

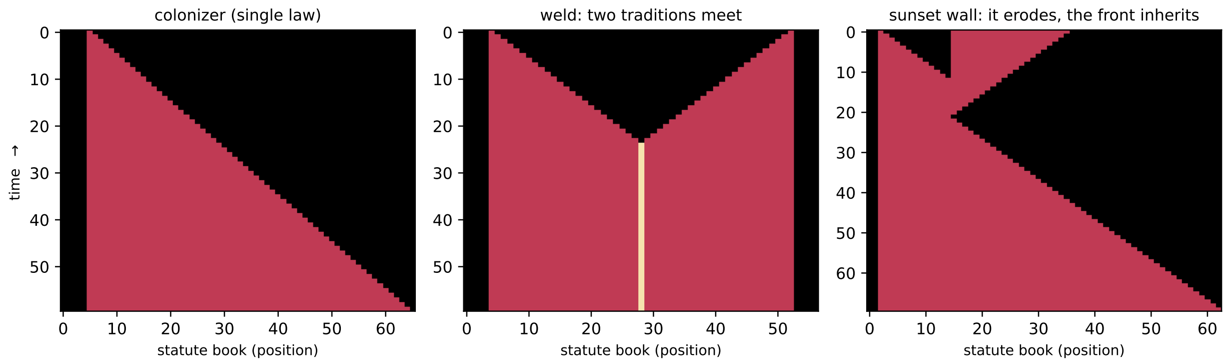

Figure 1: Window-1 nomic chains (time downward). Left: the *colonizer* $(0, 1, 1)$—"while I stand and my right is vacant, enact my kind to the right"—only its frontmost copy is ever active. Middle: two traditions meet and *weld* into frozen code with a double-law seam. Right: a colonizer meets a self-eroding *sunset wall*. The wall repeals itself from its far end while the front stands blocked; the front then fills the vacated ground, so a wall of length L is cleared at time exactly $\max(2L, x_0 + L)$ — the "refraction index 2" is those two speed-1 processes in sequence. The colonizer never converts the wall: it waits for the old code to sunset and takes the vacancy.

triple at each node—*"I am (a, b, c), and I amend the law that is (a', b', c') and amends. . . "*—and own-kind laws are exactly the self-loops. The law is still the only substance.

The expectation was that cross-kind targeting is the escape from the Anchor Theorem. It is not.

Theorem 10 (Out-Degree Law)**.** *If, restricted to the kinds a pattern uses, every law amends* at most one *kind, no free glider exists—any offsets, any window, any dimension, parity or OR. This contains the Anchor Theorem (self-loops), reciprocal amendment, and every permutation constitution of every cycle length.*

The proof is a tropical monovariant: with weights w satisfying $w_k - c_k \le w_{t(k)}$, the quantity $\Psi = \min_t(\alpha_t + w_t)$ over the kinds' trailing edges can rise only through a tight cycle of weight 0. The same argument bounds speed.

Theorem 11 (Tropical Speed Law)**.** *For a glider of period p and displacement d, $p \min(\lambda_{\min}, 0) \le d \le p \max(\lambda_{\max}, 0)$, where $\lambda_{\min}, \lambda_{\max}$ are the extreme cycle means of the amendment digraph (edge $k \to m$ of weight c_k for each $m \in T_k$). If every cycle has weight zero, or the digraph is acyclic, nothing moves.*

Theorem 12 (Supersession no-go)**.** *State-dependent* supersession *targeting—enact your own kind on empty ground, clear the whole cell if occupied—admits no free glider in any dimension: its creation channel is still own-kind, so every present kind must push forward, and then nothing can clear the rearmost cell.*

The threshold is exact: out-degree 2 suffices and the specimens are tiny (Fig. 3). The cleanest evidence is a controlled experiment—three semantics sharing their guards, their offsets and their destruction rule exactly and differing only in the out-degree of the *creation* channel—giving 0 gliders in 1,119,744 complete classifications at out-degree 1 and 3,424 certified gliders in 279,936 at out-degree 2. So *cross-amendment was never the obstruction: narrowness was.* A law that amends one provision, even somebody else's, cannot move; a law that amends two can. Entrenchment, in chapter one a consequence of linear order, is equally a consequence of legislative narrowness.

The same degree condition governs *growth*. Own-kind two-dimensional growth is pinned to axis rays ($\alpha \le 1$, a theorem of chapter one); the generalisation is exact and cheap: out-degree 1 forces $|S_t| \le |S_0|(t+1)$—hence $\alpha \le 1$ in every dimension, under parity, OR and supersession alike—while out-degree 2 attains $\alpha = 2$ from a single placed law (Fig. 4). Ray confinement was never a fact about two dimensions; it was a fact about narrow laws. The degrees of the amendment digraph turn out to govern the whole zoo: out-degree ≤ 1 forbids motion and caps growth; out-degree ≥ 2 buys gliders and area; a kind of *in-degree zero*—a provision nobody can amend—is a pump, and yields guns and rakes, on $\mathbb{Z}$ as readily as on $\mathbb{Z}^2$. *An entrenched clause is a gun.*

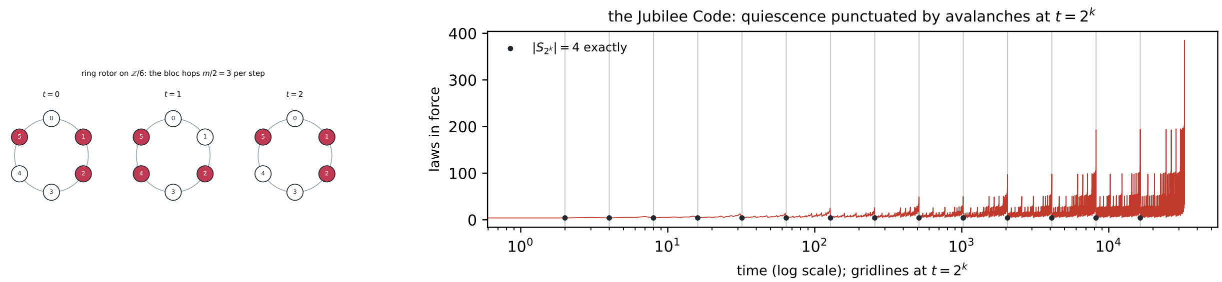

Figure 2: Left (2): the $\mathbb{Z}/6$ ring code that coincides with its own rotation by three—an apparent rotation, not transport. Right: the *Jubilee Code*, a ~26-law two-dimensional machine, aperiodic through 300,000 fully-hashed steps and quiescent except at $t = 2^k - 1$, when the counter fills and the whole code ignites; at $t = 2^k$ itself exactly four laws stand, every time (16/16, complete through $t < 2^{17}$), and the crest height doubles every second power of two. A binary counter native to law-space, and an attractor: 791 of 60,000 random seeds converge to its family.

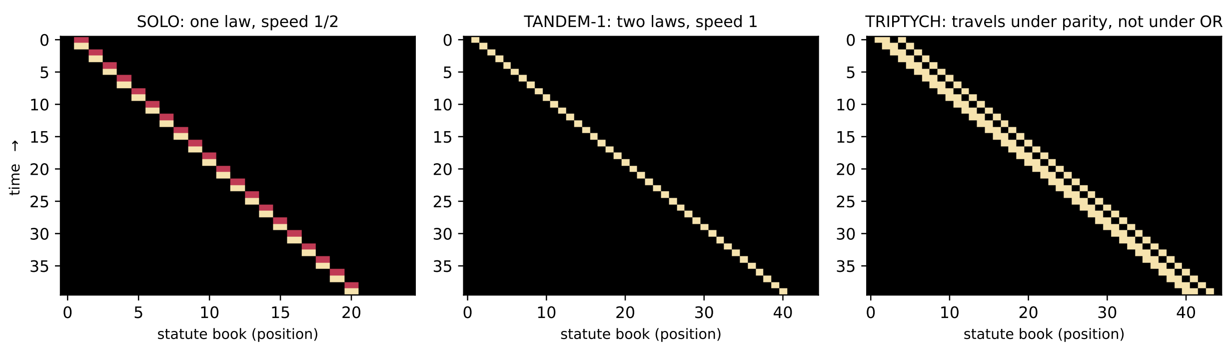

Figure 3: The first free gliders of nomodynamics (time downward). *SOLO*: one placed law travelling forever at speed $\frac{1}{2}$ under a fully-cross constitution. *TANDEM-1*: two laws in a *single* cell, period 1, speed 1—the maximum; it lifts to $\mathbb{Z}^2$ at any velocity, knight moves included, and revolves on every ring $m \geq 3$, where own-kind rotors need even $m \geq 6$. *TRIPTYCH*: a glider under parity that under OR does not travel at all but detonates into a lattice growing by three laws per step.

Cross-amendment also explains chapter one's clockwork rather than merely extending it. The frozen-occupancy step operator of an L-cycle permutation constitution lives in $\mathbb{F}_2[y]/(y^L - 1)$, *which is local exactly when L is a power of two.* So "all periods are powers of two" was never a fact about own-kind amendment; it was a fact about *cycle length one*—and 1 is a power of two. An odd factor $L' \mid L$ contributes $\mathbb{F}_{2^d}$ factors with $d = \mathrm{ord}_{L'}(2)$, and the odd periods appear at once: predicted and measured at every $L \leq 7$ ($L = 3 \to 3$, $L = 5 \to 15$, $L = 7 \to 7$), and on the three-cell ring a five-cycle constitution realises every odd period through 15 from two laws. A new resonance family appears at powers of three where own-kind gave Mersenne numbers. On rings, cross-amendment also lowers the minimal rotor circumference from six to four, reaches odd rings for the first time (always with $r \equiv \pm m/3$), and produces an object with no own-kind analogue: a *doctrinal rotor* whose occupied cells never change while the *kinds* circulate through them and return after three steps—a code frozen on its face whose doctrine rotates underneath.

Multi-authorship also revives the resolution axis that the Single-Author Lemma had made vacuous, and it kills the Dead Letter Theorem in exactly one convention.

Theorem 13 (Balance)**.** *Under parity, two active laws whose enactments cancel leave a code fixed forever while remaining alive—a* balanced constitution*; the minimal witness is two placed laws. Under OR no such code exists at any size: a fixed point still requires every law blocked, and Dead Letter survives verbatim.*

A constitution can be perpetually active and perfectly unchanging— but only where simultaneous contradictory

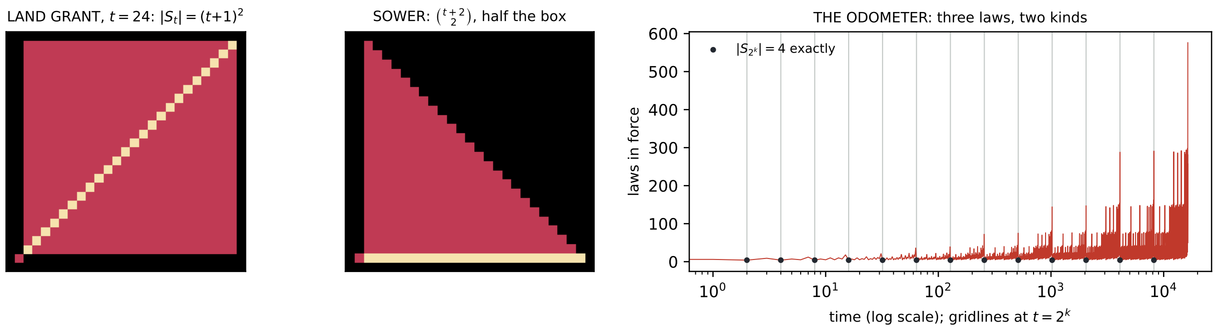

Figure 4: Two dimensions. Left: *Land Grant*—one law with two targets, and $|S_t| = (t+1)^2$ exactly, a solid growing square: the plane fills, which own-kind ray confinement forbids. Middle: *Sower*, two laws, exactly half the box. Right: *the Odometer*, three laws of two kinds, a second binary counter—quiescent, avalanching at $2^k - 1$, and standing at exactly four laws at every $t = 2^k$ (checked to $t = 2^{20}$), with reach growing like $\sqrt{t}$ (measured ratio 1.500 at every even power of two, to $t = 2^{17}$): bounded population, unbounded memory, never repeating.

amendments annul one another instead of both taking effect. Balance is governed by the *other* degree: a provision with two authors. It is not a curiosity of small codes—balanced codes of unbounded size exist with *every* law active at every step, their exact count satisfies $a(s) = 4a(s-1) - 2a(s-2)$, and 70% of the balanced verdicts in a complete 143-million-run census are reached rather than seeded.

Finally, the Anchor Theorem itself has a deeper reading than "own-kind amendment forbids motion". A sweep of the semantics the referent suggests—entrenchment clauses, quorum guards, later-law-wins, sunset-by-default—finds that all but one leave the confinement theorems untouched (they are guard-free), while *sunset-by-default*, where a law lapses unless re-enacted, admits free gliders on $\mathbb{Z}$ with plain own-kind targeting: 11–15% of a complete 139,968-code box, at speeds 1, $\frac{2}{3}$, $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{6}$. So there are exactly two ways known to make a statute book move:

legislate broadly (out-degree ≥ 2), or let provisions expire on their own.

What forbids motion on the line is **permanence**, not own-kind targeting—the Anchor Theorem's real content.

6 Impermanence: motion is a consequence of mortality

Every sector so far assumes a law stands until something repeals it. That assumption is load-bearing and nobody had weighed it. Drop it: take the lifetime τ of §2.3 to be finite, so that effects are enactments and a placed law survives only while it keeps being re-enacted. To repeal is then simply to stop re-enacting, and $\tau = 1$ is the pure case: what stands next is exactly what is enacted now.

Theorem 14 (Lone Survivor)**.** *Under $\tau = 1$ a single placed law of kind (a, b, c) survives its first step iff $a = 0$ and $b \neq 0$—its precedent is itself and its exception is elsewhere. It then translates by c forever: it* walks *if $c \neq 0$ and renews itself in place if $c = 0$. The other 21 of the 27 kinds vanish in one step.*

So chapter one's very first specimen becomes a glider (Fig. 5), and the Anchor Theorem's locality hypothesis—*a slot receiving no toggle is unchanged*—is revealed as permanence itself. Structurally the term is visible in the monovariant: $\mathrm{supp}_t(n+1) \subseteq \mathrm{supp}_t(n) \cup (\mathrm{supp}_s(n) + c_s)$, whose first summand is a **weight-zero self-loop** in the tropical digraph—exactly what pins the minimum. Impermanence deletes it, the cycle means may go positive, and motion is permitted. The statistics invert with it: in the complete two-law box (3,645 codes per lifetime) extinction becomes the generic fate ($\approx 64\%$) where gridlock had been, and gliders—*impossible* in the whole own-kind permanent sector—occur in $\approx 23\%$ of codes at every lifetime tested.

Gridlock inverts rather than merely failing. A solid block is frozen under permanence because every vacancy guard fails; under impermanence it *evaporates*, since nothing re-enacts the interior, and eight colonizer

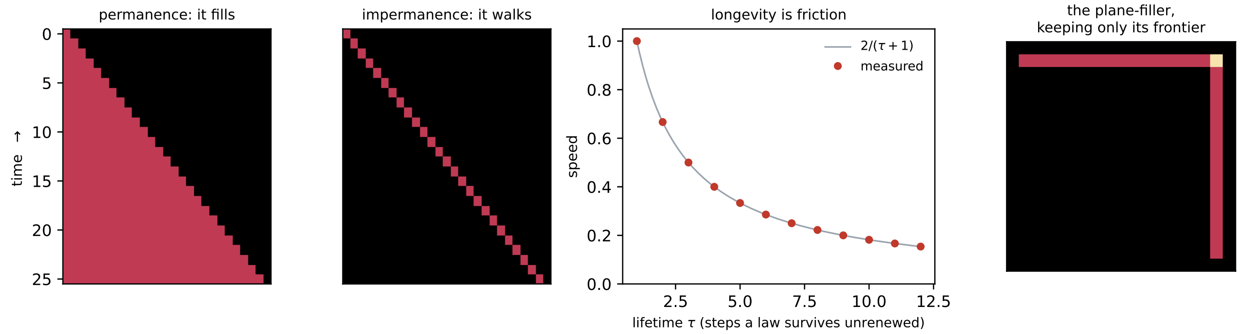

Figure 5: Impermanence. The colonizer $(0, 1, 1)$—the same law, the same guard, the same target—*fills* the line when provisions are permanent and *walks* when they lapse. Third panel: one code, one seed, and the lifetime τ as a dial; the packet always moves two cells but must wait for its own tail to lapse, so its speed is exactly $2/(\tau + 1)$ (measured points, certified $\tau \leq 12$). Right: *Land Grant*, whose population is $(t + 1)^2$ under permanence, keeps only the two advancing edges of that square.

laws collapse in one step to a single walker. *In both worlds only the surface matters: in one it is the only part that moves, in the other the only part that lives.*

Theorem 15 (Conservation)**.** *Under $\tau = 1$, if every law amends at most one kind then $|S_{n+1}| \leq |\mathrm{active}(S_n)| \leq |S_n|$: an impermanent code can never grow.*

Proof. The next state is exactly the set of slots enacted now; each active law enacts one, and nothing else survives. □ With out-degree 2 growth appears at once. So one threshold governs both worlds with dual consequences: out-degree ≤ 1 forbids *motion* under permanence and *growth* under impermanence; out-degree ≥ 2 buys each.

7 The statute book computes

Three chapters were spent on motion—proving it impossible, pricing it, and finding that it requires mortality. Computation turns out to need none of it.

A guard reads *some law stands at $i + a$ and none stands at $i + b$*: an AND-NOT at fixed offsets. Parity resolution makes several authors of one provision XOR together. AND-NOT with XOR is functionally complete, and the one missing ingredient—*assignment*, since toggles accumulate—is supplied by a single observation.

Lemma 16 (the self-clearing kind)**.** *Let a kind v have rule $(0, 0, 0)$, a vacuously true guard and target $\{v\}$. A v-law is then active whenever it stands, and it toggles itself. Hence, with $f(t)$ the XOR of every other author's toggle of that slot, $x(t + 1) = x(t) \oplus x(t) \oplus f(t) = f(t)$.*

A provision that expires every step is a register: the code writes it rather than amending it. A law nobody amends is a **gate**; a law that repeals itself is a **wire**.

Theorem 17 (Statute–Circuit)**.** *A constitution in the resulting normal form* is *a synchronous AND-NOT network with free XOR fan-in, free fan-out (guards are read-only, so any number of gates may cite one slot) and unit delay.*

Theorem 18 (universality)**.** *Rule 110 runs inside a* 24*-kind window-*1 *constitution at three amendment steps per Rule-110 step. Moreover every Turing machine compiles into a* finite code *of the founding occupancy-guard sector, its tape supplied by a self-extending front; so nomodynamics is computation-universal and halting is undecidable for finite codes.*

The Rule-110 certificate is complete rather than sampled: every offset lies within ± 1, so three steps form a local map on a seven-cell window, and running all 2^7 configurations of $\mathbb{Z}/7$ evaluates that map at every one of its inputs, which decides $\mathbb{Z}$ entirely. Prediction is **P-complete** already at dimension 1, window 0, one cell; by contrast the single-author sector is $\mathbb{F}_2$-linear with $\Phi^t = L^t$, so prediction *there* lies in NC and is not P-complete unless NC = P. Chapter one's exactly solvable sector is provably the wrong place to look for computation—which is precisely why it was solvable.

Two things this does *not* say. Nothing here is minimal: 24 kinds is what one construction needed. And the ballistic route—Life-style glider collisions—does not work: 58,203 certified collisions produced no reflector, no fan-out, no transparency and no annihilation, one glider family freezing everything it touches and another detonating everything it touches, both gateless for opposite reasons. Every such zero is a statement about a box.

What the construction does say is worth stating plainly. Every gate law and every power law sits exactly where it started, certified over 60 steps at out-degree 1:

A statute book can compute while every one of its provisions stands still.

What moves is not law but information, and information moves because guards read across cells. *Motion was never the resource; the vacancy clause was.*

8 Replication

Von Neumann's rule-carrying constructor is one of the three fragments this field was derived from, so the question is obligatory. Codes do copy themselves. The Split Decision (two kinds, two dimensions, citation, OR) performs true binary fission—$\Phi^2(S) = \sigma^{(0,-2)}(S) \sqcup \sigma^{(0,+2)}(S)$ exactly, the parent not surviving, the children beyond interaction range, the debris empty—and at every even t the state is exactly $t/2+1$ free exact copies. The Engrossment needs no citation at all: plain occupancy guards and parity, the founding semantics, with $\Phi^4(S) = S \sqcup \sigma^{(4,4)}(S)$ and $2^{\mathrm{popcount}(t/4)}$ copies.

Two results shape the sector. First, replication here is *not* the known additive phenomenon of $\mathbb{F}_2$-linear automata: the replicators fail the splitting test—remove one law of four, evolve the halves separately, XOR, and the child is simply absent—while the genuinely additive specimens are reported as Fredkin and nothing more. (A caution found the hard way: $2^{\mathrm{popcount}(t)}$ is *not* a signature of additivity; it survives into demonstrably non-additive systems.) Second, the light cone bounds the population: $\mathrm{card}(S_t) \le n(s_0 + 2Rt + 1)^D$, so **exponential replication is impossible in every dimension** and no fixed-period doubling can survive—every free fission must eventually collide with its own descendants, and The Engrossment accordingly doubles at $t = 4(2^k - 1)$, at exponentially stretching intervals, exactly as forced.

Von Neumann's full constructor was not reached, but both of its halves were: a code that reads a blueprint carried in its own body and builds a decoded, different target from it, and a code that copies an arbitrary blueprint faithfully without bound. Neither ever holds two complete copies of its machinery at once, and the obstruction is structural—a child built next door lies within the interaction radius of its parent, so it can become free only by travelling. Hence the sharpest open problem the subject now has: *a constructor in nomodynamics is a blueprint-carrying packet that moves.* Travelling needs out-degree ≥ 2; reading needs citation; nothing found so far forbids the combination.

9 The frontier, located by theorem

The theorems above are cheap to state and were found within two days of the definition, which is the usual sign that a territory has been entered rather than exhausted. Open problems, in the order we would attack them. (1) *Speed quantisation, and a warning about boxes.* The apparent law "the number of kinds caps the speed" is an artefact, and the way it failed is worth more than the law would have been. Bounded searches decide a *box*; every census in this program used boxes of at most ∼26 interior cells, and the two-kind universe Mirror ($W = 1$, rules $(0, 1, -1)$ and $(0, -1, 1)$, each amending both kinds) carries displacement-2 gliders of minimal period $4, 5, 6, 7, 12$ with spans 53, 20, 616, 438, 39—all invisible to those boxes, and all certified. A second confound compounds it: sweeps over *coprime* (p, d) are structurally blind to $\Phi^4 = \sigma^2$ when $\Phi^2 = \sigma^1$ fails. So every "decided impossible" in a fixed box, here and elsewhere, means *no narrow glider.* What does survive is sharper: in the single-field sector (all kinds sharing the same target set, so the system is a one-bit automaton with n channels) the measured cap is $|d| \le 2$ under parity and $|d| \le 1$ under OR, *at any width and any number of channels*; the resource that buys $|d| \ge 3$ is another $\mathbb{F}_2$ field, not another kind. Prove that cap, and explain why it fails to scale with W: at $W = 2$ and three kinds displacements reach at least $5 > 2W$, while a dilation $(W, p, d) \mapsto (rW, p, rd)$ rules out any width-independent bound a priori. (2) Is any small cross-amendment universe computation-universal? The gate-level inventory—what reflects, what annihilates, what survives collision—comes first; the claim comes much later, if at all. (3) Is there a *light-cone-admissible* rotor on any odd ring—one that actually carries something round—or is every odd-ring witness a barber pole? And can the block/gap calculus be pushed past blocks of length two, which would decide the own-kind odd rings outright instead of up to $m = 31$? (4) Prove the Jubilee clock law—$|S_t| = 4$ at *every* power of two, crest at $2^k - 1$ doubling every second power—and explain its basin: 791 of 60,000 random seeds fall into this one attractor. (5) The reception question the founding practice suggests: which theorems survive into semantics faithful to actual amendment procedure—entrenchment clauses, quorum guards, later-law-wins, sunset by default?

Methods. Definition and predictions were frozen before the first run; censuses carry machine-checked certificates (recurrence, translation-recurrence, rotation-recurrence); complete enumerations: 1- and 2-law seeds (W=1), $\leq$4-law/5-cell patterns (8.1M), ring sweeps $m \leq 12$; the Anchor and Dead Letter theorems have short combinatorial proofs checked against 15.3M runs with zero violations. Full records: the Treeline program archive, `phase6/`.